

PAPER_1515_TRANSCENDENTAL_LN_10

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PAPER_1515 — $\ln 10 = (1+F_TRZ)(K_MEX + F_TRZ^2)$ — Residual 0.0035%

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket A (Foundational mathematical identity)

Date: June 18, 2026

Status: CLOSED — Natural logarithm of 10 expressible in UQFF primitives at sub-0.01% precision

Observation

PAPER_1208 (UQFF Transcendentals Unified Proof Set) demonstrates that the natural logarithm $\ln 10 \approx 2.302585$ can be expressed using only the locked UQFF integer primitives F_TRZ and K_MEX , with no fitted constants.

UQFF Closed Identity

$$\begin{aligned}\ln 10 &\approx (1 + F_TRZ) \cdot (K_MEX + F_TRZ^2) \\ &= (1 + 1/10) \cdot (25/12 + 1/100) \\ &= 1.1 \cdot 2.09333... \\ &= 2.30267\end{aligned}$$

Observed: $\ln 10 = 2.302585093...$

Residual: $|2.30267 - 2.30259| / 2.30259 = 0.0035\%$

Equivalent Forms

The expression factors elegantly:

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$$\begin{aligned}\ln 10 &= K_MEX + F_TRZ \cdot K_MEX + F_TRZ^2 + F_TRZ^3 \text{ (raw expansion)} \\ &= (1 + F_TRZ)(K_MEX + F_TRZ^2) \text{ (factored form)} \\ &= K_MEX(1 + F_TRZ) + F_TRZ^2(1 + F_TRZ) \text{ (mixed)}\end{aligned}$$

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Each form involves only two integer primitives ($F_TRZ = 1/10$, $K_MEX = 25/12$) — no additional fitted parameters.

Significance

This identity demonstrates that the UQFF integer primitives are **not arbitrary physics-fitted values** but encode rational approximations to fundamental mathematical constants. The natural logarithm of 10, which appears in:

- Cosmology (orders of magnitude of physical scales)

- Decibel/pH/Richter scales
- Information theory (digit count)
- Stirling approximations

...emerges to 0.0035% precision from $\{F_{\text{TRZ}}, K_{\text{MEX}}\}$ alone.

NOT REPLACEMENT

This is not a claim that $\ln 10$ IS equal to $(1+F_{\text{TRZ}})(K_{\text{MEX}}+F_{\text{TRZ}}^2)$. It is a claim that the UQFF integer primitives carry a structural rational approximation to $\ln 10$ with sub-percent precision, demonstrating their mathematical depth.

Reference

- Source: PAPER_1208 UQFF Transcendentals Unified Proof Set
- Related: PAPER_1516 ($\ln 2$), PAPER_1517 (π^2)
- Calculator dispatch: `calculate_paradox({"paradox": "transcendental_ln_10"})`

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